Package ‘aocseq’

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Type Package

Title aocseq 0.1.0: An R package for mixed cell type annotation following the activation of cells & sequencing.

Version 0.1.0

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Description aocseq is a suite of statistical tools for the analysis of multimodal immunosequencing data. The purpose of this statistical tool is to quantify single cell data and provide 1) A sample list of mixed cell types with accompanying data sheets to measure the number of different cell types that are present in the blood or tissue and 2) To detect, score and rank cells with optimal or desirable characteristics. Characteristics are defined by the gene expression of cells that are considered of interest because of their response to perturbation.

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Encoding UTF-8

LazyData true

Imports Seurat, matrixStats, readr, Rcpp, stats, BiocParallel, Matrix, MASS, VGAM, bbmle, gamlss, maxLik, pscl, rjson

RoxygenNote 7.3.2

R topics documented:

`AnnotateCellTypes` *List cell types and the proportion of cells expressing a choice of marker genes at normalized counts greater than the control*

Description

This function imports a Seurat object list processed by the aocseq::CombineData function and produces a spreadsheet of cell types raned by frequency. The table displays the percentage of CD4 and CD8 cells expressing a choice of marker genes at normalized counts greater than the control.

Usage

```
AnnotateCellTypes(
  cell.data,
  goi,
  cell.type = "cdr3_na",
  goi.threshold = 0.975,
  TCR = TRUE,
  index.control.input = 1,
  conditions = 1,
  path = "",
  names.spreadsheet = c(-1),
  preset = 1,
  threshold.entry = FALSE
)
```

Arguments

`cell.data`	A Seurat object pre-processed with aocseq::CombineData.
`goi`	A marker of interest.
`cell.type`	Clone identifier, e.g. cdr3_na, phenotype.
`goi.threshold`	Quantile that determines high counts per cell.
`TCR`	True or false if true, adds amino acid sequence of TCR if cell.type is set to cdr3_na.
`index.control.input`	Index of the control sequencing sample for each condition.
`conditions`	Number of experiments within cell.data object.
`path`	String. Directory of the table output.
`names.spreadsheet`	List. Control and sample names.

`preset`	Bool. If set to 1, the control is part of the dataset. If 0, a threshold value for each condition must be set.
`threshold.entry`	List. List of thresholds for each condition if preset is 0.

Value

A data frame containing clonotypes and summary statistics of transcript expression.

`CellTypeLoop` *Return a data frame of meta data associated with single cell data sets that contains a distance from a reference that is determined using the isolation forest algorithm*

Description

This function will return the percentage of cells that are outliers when compared to a reference data set.

Usage

```
CellTypeLoop(
  cell.data,
  cell.types,
  reference.data,
  genes,
  ntrees,
  maxheight,
  solver = TRUE,
  subsample.count = ncol(reference.data) + 1
)
```

Arguments

`cell.data`	A Seurat object containing cells that are to be assigned a distance.
`cell.types`	A list of the cell types (currently this is clonotypes).
`reference.data`	Normalized reference data.
`genes`	Genes to pull from the query data set.
`ntrees`	Number of trees used for the isolation forest.
`maxheight`	Maximum height used for the isolation forest.
`solver`	True or false If true an Rcpp implementation of the software is used
`subsample.count`	Sub sampling number for isolation forest.

Value

A data frame containing the mean normalized isolation forest score for each cell type.

ClassifyCells *Cell classifier function that assigns a mean distance per cell type based on several metrics used for single cell RNA sequencing classification.*

Description

This function will compute a classification for each single cell in samples that have been activated and sequenced. Takes as input two Seurat objects, one containing the query cells and another containing the reference. A gene list used to construct the distance and output path for plotting and storing the mean distance per cell type. Other parameters are listed for debugging, but can be left as default values.

Usage

```
ClassifyCells(
  output.array,
  reference.data,
  gene.list,
  cell.types = c(),
  distance = 0,
  scramble = FALSE,
  withSCT = FALSE,
  ntrees = 10,
  maxheight = 20,
  PRINT = FALSE,
  save.dir = ".",
  verbose = TRUE
)
```

Arguments

`output.array`	A Seurat object containing cells that are to be assigned a score.
`reference.data`	Matrix. Reference data used to score cells.
`gene.list`	Vector. List of genes in the gene signature.
`cell.types`	Vector. List of cell types for classification.
`distance`	Can be set to 0,1,2 or 3. Choice of 0 returns the Mahalanobis distance, 1 returns the taxicab distance, 2 returns the z-score and 3 returns the isolation forest distance.
`scramble`	Bool. To asses the effect of the gene signature on the Rscore, select a random set of genes and compute their distance from the reference data.
`withSCT`	True or False. Default=False. If TRUE, classification is made on sctransform data.
`ntrees`	Number of trees used for the isolation forest algorithm.
`maxheight`	Maximum height of the tree used for the isolation forest algorithm.
`PRINT`	True or false. If true results will be saved to save.dir.
`save.dir`	path to save metadata as spreadsheet if PRINT is set to TRUE.
`verbose`	Print progress bars and output

Value

A Seurat object containing aocseq cell classification metadata.

ClassifyCellTypes — *Combine single cell data and annotate with cell labels based on functionality*

Description

This function will read in Seurat objects that have been processed with aocseq, and accompanying annotation tables to classify cell types listed in annotation.tab using Bayesian inference and distance scores. Takes as input gene expression from scRNAseq and VDJ enrichment. Outputs a classification table that lists samples and each clonotype or cell type and the classification.

Usage

```
ClassifyCellTypes(
  cell.data,
  annotation.tab,
  cell.type = "cdr3_na",
  within.sample = FALSE,
  method = "ZINB",
  goi = c("IFNG"),
  percentile = 0.01,
  path = "",
  path.glist = "",
  reference = matrix(nrow = 1, ncol = 1),
  verbose = TRUE
)
```

Arguments

`cell.data`	A seurat object. Pre-processed with aocseq.
`annotation.tab`	Table. An annotation table from the aocseq package.
`cell.type`	Cell types to classify each sub population.
`within.sample`	True or False. Set to TRUE for comparison between control and test data. For within sample cell type annotation, set to FALSE.
`method`	Choice of "ZINB", "Mahalanobis", "z-score", "IsoForest" and "Taxi cab". "ZINB" can be used without a reference data set.
`goi`	Gene. Gene of interest (goi). For use with "ZINB" classification.
`percentile`	Double. Percentile cutoff for distance based scoreing metrics.
`path`	Directory for output tables.
`verbose`	Print progress bars and output.

Value

A Seurat object list containing metadata and VDJ annotations.

`CombineData` *Combine single cell data and annotate with cell labels based on functionality*

Description

This function will read in 10X genomics outputs from cellranger and combine multiple Seurat objects, assigning labels based on additional tables stored in csv format. Takes as input gene expression from scRNAseq and VDJ enrichment. A gene list used to estimate specificity can be chosen and is application specific. Other parameters are listed for debugging, but can be left as default values.

Usage

```
CombineData(
  gex.path,
  marker.gene,
  vdj.path = c(),
  threshold.cutoff = 0.975,
  file.saved = "samples.rds",
  index.control = c(-1),
  sample.name = c(-1),
  preset = 1,
  threshold.entry = 0,
  demultiplex = FALSE,
  demultiplex.index = c(),
  nameshashtags = c(),
  n.ht.per.sample = 1,
  tenX_conversion = "true",
  nFeature_RNA_lower = 100,
  nFeature_RNA_upper = 10000,
  nvariable_features = 3000,
  percent.mt_upper = 5,
  data.input = "raw",
  upperQ = 0.95,
  lowerQ = 0.05,
  verbose = TRUE,
  QC_plots = FALSE
)
```

Arguments

`gex.path`	path to gene expression data in the format of cellranger feature barcode matrices.
`marker.gene`	list of marker genes used for specificity analysis.
`vdj.path`	path to VDJ expression data in the format of cellranger csv files.
`threshold.cutoff`	list of marker genes used for specificity analysis.
`file.saved`	directory for storing Seurat object RDS files.
`index.control`	Index of the control sequencing sample.

`sample.name`	Names that will be added to each sample orig.ident.
`preset`	If set to 1, preset determines the threshold for cutoff if a control is included in the assay. If set to 0 a cutoff is set by the threshold.entry, which defines what value is high for all samples. If set to 2 a cutoff is set per sample.
`threshold.entry`	Double. Sets the percentile above which gene expression is labelled as high.
`demultiplex`	Boolean value used to indicate if the data was hashtagged and therefore requires to be split.
`demultiplex.index`	Indexes to select columns with hashtag counts in the 10X genomics Seurat object assay. This depends on the 10X genomics chemistry and output from cellranger.
`n.ht.per.sample`	Number of hashtags used per sample (only set if hashtagging has been used).
`tenX_conversion`	Parameter for 10X compatability.
`nFeature_RNA_lower`	Threshold for the minimum number of unique gene identifiers detected in a single cell.
`nFeature_RNA_upper`	Threshold for the maximum number of unique gene identifiers detected in a single cell.
`nvariable_features`	Threshold for the maximum total number of unique gene identifiers detected in a single cell for dimension reduction.
`percent.mt_upper`	Threshold for the maximum percentage of unique mitochondrial gene identifiers detected in a single cell.
`verbose`	Print progress bars and output.
`QC_plots`	Print progress bars and output.
`names.hashtag`	hashtag sample names.

Value

A Seurat object list containing metadata and VDJ annotations.

`ConstructTree` *Build a binary tree from a numerical data frame*

Description

This function will build a binary tree from a numerical data frame and return a data frame containing each unique data point from the input data and the height at which each data point becomes "isolated" (reaches an external node) or the max_height.

Usage

```
ConstructTree(data, current_height, max_height, kurtosis = TRUE)
```

Arguments

`data`	a numerical data frame.
`current_height`	A parameter which tracks the current height of a given construct_tree instance, allows for recursive calling of the function.
`max_height`	The maximum height the tree will grow to before stopping
`kurtosis`	True or False. If set to TRUE, kurtosis splits the data based on the kurtosis over the maximum gene distribution across the cells in the input data.

Value

A data frame containing the data for each unique point in the input data and the height at which that point was isolated.

`DEsingle` *This is a copy of DEsingle that was lifted from the orginal package, see DOI: 10.18129/B9.bioc.DEsingle It is included in the aocseq package with modifications for package compatibility. For information on the function, see the comment pasted from the tool. DEsingle: Detecting differentially expressed genes from scRNA-seq data This function is used to detect differentially expressed genes between two specified groups of cells in a raw read counts matrix of single-cell RNA-seq (scRNA-seq) data. It takes a non-negative integer matrix of scRNA-seq raw read counts or a* `SingleCellExperiment` *object as input. So users should map the reads (obtained from sequencing libraries of the samples) to the corresponding genome and count the reads mapped to each gene according to the gene annotation to get the raw read counts matrix in advance.*

Description

This is a copy of DEsingle that was lifted from the orginal package, see DOI: 10.18129/B9.bioc.DEsingle It is included in the aocseq package with modifications for package compatibility. For information on the function, see the comment pasted from the tool. DEsingle: Detecting differentially expressed genes from scRNA-seq data This function is used to detect differentially expressed genes between two specified groups of cells in a raw read counts matrix of single-cell RNA-seq (scRNA-seq) data. It takes a non-negative integer matrix of scRNA-seq raw read counts or a `SingleCellExperiment` object as input. So users should map the reads (obtained from sequencing libraries of the samples) to the corresponding genome and count the reads mapped to each gene according to the gene annotation to get the raw read counts matrix in advance.

Usage

```
DEsingle(counts, group, goi, parallel = FALSE, BPPARAM = bpparam())
```

Arguments

`counts`	A non-negative integer matrix of scRNA-seq raw read counts or a `SingleCellExperiment` object which contains the read counts matrix. The rows of the matrix are genes and columns are samples/cells.

`group`	A vector of factor which specifies the two groups to be compared, corresponding to the columns in the counts matrix.
`parallel`	If FALSE (default), no parallel computation is used; if TRUE, parallel computation using `BiocParallel`, with argument `BPPARAM`.
`BPPARAM`	An optional parameter object passed internally to `bplapply` when `parallel=TRUE`. If not specified, `bpparam()` (default) will be used.

Value

A data frame containing the differential expression (DE) analysis results, rows are genes and columns contain the following items:

- theta_1, theta_2, mu_1, mu_2, size_1, size_2, prob_1, prob_2: MLE of the zero-inflated negative binomial distribution's parameters of group 1 and group 2.
- total_mean_1, total_mean_2: Mean of read counts of group 1 and group 2.
- foldChange: total_mean_1/total_mean_2.
- norm_total_mean_1, norm_total_mean_2: Mean of normalized read counts of group 1 and group 2.
- norm_foldChange: norm_total_mean_1/norm_total_mean_2.
- chi2LR1: Chi-square statistic for hypothesis testing of H0.
- pvalue_LR2: P value of hypothesis testing of H20 (Used to determine the type of a DE gene).
- pvalue_LR3: P value of hypothesis testing of H30 (Used to determine the type of a DE gene).
- FDR_LR2: Adjusted P value of pvalue_LR2 using Benjamini & Hochberg's method (Used to determine the type of a DE gene).
- FDR_LR3: Adjusted P value of pvalue_LR3 using Benjamini & Hochberg's method (Used to determine the type of a DE gene).
- pvalue: P value of hypothesis testing of H0 (Used to determine whether a gene is a DE gene).
- pvalue.adj.FDR: Adjusted P value of H0's pvalue using Benjamini & Hochberg's method (Used to determine whether a gene is a DE gene).
- Remark: Record of abnormal program information.

Author(s)

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See Also

`DEtype`, for the classification of differentially expressed genes found by `DEsingle`.

`TestData`, a test dataset for DEsingle.

Examples

```
# Load test data for DEsingle
data(TestData)

# Specifying the two groups to be compared
# The sample number in group 1 and group 2 is 50 and 100 respectively
group <- factor(c(rep(1,50), rep(2,100)))

# Detecting the differentially expressed genes
```

```
results <- DEsingle(counts = counts, group = group)

# Dividing the differentially expressed genes into 3 categories
results.classified <- DEtype(results = results, threshold = 0.05)
```

EStep *EStep*

Description

EStep

Usage

```
EStep(x, mu.vector, sd.vector, alpha.vector)
```

Arguments

sd.vector	Vector containing the mean of each component.
alpha.vector	Vector containing the mixing weights of each component.

Value

Named list containing the loglik and posterior.df.

GetGeneSignature *This function saves differential gene expression from multiple subsets from samples that have been processed and classified by aocseq.*

Description

This function will take a set of annotation spreadsheets and export differential expression between different cell types labelled by their marker genes Currently the phenotypic separation is CD4 and CD8 T cell phenotypes, but the user is encouraged to generate as many subsets as needed for downstream analysis.

Usage

```
GetGeneSignature(
  cell.data,
  annotation.path,
  save.dir,
  goi,
  FClim = 0,
  specific.pval = 10^(-5),
  bystander.pval = 10^(-2),
  set.min.pct = 0.25,
  logfc.threshold = 0
)
```

Arguments

`cell.data`	A Seurat object pre-processed with aocseq::CombineData.
`annotation.path`	Character array. Directory of an aocseq clonotype annotation table.
`save.dir`	Directory for storing differential genes.
`goi`	Character array. Gene name, a marker of interest.
`FClim`	Fold change cutoff for differential genes.
`specific.pval`	If cells are classified by a zero inflated negative binomial (ZINB) model, sets the significance cutoff for specificity in each cell subset specified by the cell type metadata.
`bystander.pval`	Same parameter as used for specific.pval, but sets the threshold for bystanders.
`set.min.pct`	Same set.min.pct in Seurat FindMarkers function.
`logfc.threshold`	Same logfc.threshold in Seurat FindMarkers function.

Value

return an assay containing predicted expression value in the data slot

---

`GetSpecificCells` *Subsets a SEURAT object based on a list of meta data for the CD4 and CD8 subsets*

Description

This function will read in Seurat objects processed by aocseq::AnnotateCellTypes or any other software based on cell annotation and subset the array accordingly

Usage

```
GetSpecificCells(cell.data, expression, phenotype, cell.types, goi_v)
```

Arguments

`cell.data`	A Seurat object containing single cell RNA sequencing data.
`expression`	Choice of "high" or "unassigned".
`phenotype`	CD4 or CD8 classification.
`cell.types`	List of cell types to subset single cells included in the analysis.
`goi_v`	Gene of interest.

Value

A subset of a Seurat object.

GetTCR — *Code to pull the full length TCR sequence from 10X genomics all_contig_annotations.json file*

Description

This function will import a set of single cell barcodes, extract the full length TCR transcript details and output the details ranked by the cell with the greatest functional score.

Usage

```
GetTCR(
  barcodes,
  contig.annoations,
  json.path = ".",
  save.dir = ".",
  score = "",
  verbose = TRUE
)
```

Arguments

barcodes	Barcodes used to identify cells in full contig annotations JSON file.
contig.annoations	contig_annotations.csv file.
json.path	Directory where JSON file is saved.
save.dir	Directory where full length TCR transcripts will be saved.
score	Orders the cells be classification score.
verbose	Print progress bars and output

Value

Void

GMMDemux — *Flexible implementation of the GMM demux algorithm*

Description

Demultiplex arbitrary number of hashtags

Usage

```
GMMDemux(
  s.name,
  data,
  hashtag.index,
  nameshashtags,
  set.col.name = "Hashtags",
  Seurat_Object = FALSE
)
```

Arguments

`s.name`	This function creates a 10X genomics Seurat object. s.name sets the name of the project.
`data`	Input data. 10x genomics Seurat object with hashtag antibodies.
`hashtag.index`	List containing positions of hashtags to be de-multiplexed.
`nameshashtags`	Sets the values of the sample for the new orig.ident to be added to the de-multiplexed Seurat object.
`set.col.name`	Sets the column name to be added to the de-multiplexed Seurat object.
`Seurat_Object`	True or false. Default is FALSE and assumes input data is a data frame or matrix.

Value

A Seurat object that has been split by hastagged demultiplexing.

`IsoForest` *Build many trees on samples from data and average the heights*

Description

Build many trees on samples from data and average the heights

Usage

```
IsoForest(
  data,
  num_trees,
  max_height,
  subsample_count = nrow(data),
  kurtosis_param = TRUE
)
```

Arguments

`data`	A numerical data frame.
`num_trees`	The number of trees to build and average over.
`max_height`	The maximum height each tree will grow to before stopping.
`subsample_count`	The size of the random sample that a tree will be built on. Cannot be larger than the number of rows in the data.
`kurtosis_param`	True or False. If set to TRUE, kurtosis_param splits the data based on the kurtosis over the maximum gene distribution across the cells in the input data.

Value

A data frame containing the data for each unique point in the input data and the average height at which that point was isolated.

MakeReference — *These are functions to produce reference datasets to compute a response score of activated or perturbed T cells.*

Description

This function returns a matrix of normalized gene expression along the rows and cells along the columns.

Usage

```
MakeReference(
  cell.data,
  gene.list,
  cell.type.list,
  celltype = cdr3_na,
  threshold = 0.975,
  n.inlist = 10,
  cellcycle = FALSE,
  SCT = FALSE,
  verbose = TRUE
)
```

Arguments

`cell.data`	List of Seurat objects. Seurat objects containing expression data processed. by aocseq that will be used to score query T cell response.
`gene.list`	List of genes to be included in the reference matrix.
`cell.type.list`	Character list. List of cell types to include in the reference data.
`celltype`	Cell type to access metadata of reference dataset.
`n.inlist`	Number of genes with high expression in each cell included in the reference matrix.
`cellcycle`	Adds Seurat cell cycle phase to reference data.
`SCT`	Returns reference matrix in sctransform space.
`verbose`	Print progress bars and output.

Value

A reference matrix.

MStep *Maximization Step of the EM Algorithm*

Description

Update the Component Parameters. Iterative function for the expectation maximization (EM) algorithm.

Usage

```
MStep(x, posterior.df)
```

Arguments

`x`	Input data.
`posterior.df`	Posterior probability data.frame.

Value

Named list containing the mean (mu), variance (var), and mixing weights (alpha) for each component.

NormalizationScore *This function normalizes the data return by the aocseq R implementation of the isolation forest algorithm.*

Description

This function normalizes the data return by the aocseq R implementation of the isolation forest algorithm.

Usage

```
NormalizationScore(df)
```

Arguments

`df`	A data frame containing isolation forest distances.

Value

Normalized values.

`PercentOutlier` *Return a data frame of meta data associated with single cell data sets that contains a distance from a reference.*

Description

This function will return the percentage of cells that are outliers when compared to a reference data set.

Usage

```
PercentOutlier(
  cell.data,
  reference.data,
  genes,
  ntrees,
  maxheight,
  subsample.count = ncol(reference.data) + 1
)
```

Arguments

`cell.data`	A Seurat object containing cells that are to be assigned a distance.
`reference.data`	Normalized reference data.
`genes`	Genes to pull from the query dataset.
`ntrees`	Number of trees used for the isolation forest.
`maxheight`	Maximum height used for the isolation forest.
`subsample.count`	Subsampling number for isolation forest.

Value

A data frame with the percentage of cells that are outliers.

`QCPlot` *QC plot*

Description

This function will import raw feature barcode matrix and plot the number of genes detected per cell, the number of reads detected per cell and the percentage of mitochondrial genes per cell. The results are displayed as QC thresholds to vizualize the cells that are removed from the data.

Usage

```
QCPlot(
  cell.data,
  nFeature_RNA_lower = 100,
  nFeature_RNA_upper = 10000,
  nCount_RNA_lower = 100,
  nCount_RNA_upper = 10000,
  percent.mt_upper = 5
)
```

Arguments

`cell.data` A Seurat object pre-processed with aocseq::CombineData.

`nFeature_RNA_lower`
Threshold for the minimum number of unique gene identifiers detected in a single cell.

`nFeature_RNA_upper`
Threshold for the maximum number of unique gene identifiers detected in a single cell.

`nCount_RNA_lower`
Threshold for the minimum number of total RNA reads detected in a single cell.

`nCount_RNA_upper`
Threshold for the maximum number of total RNA reads detected in a single cell.

`percent.mt_upper`
Threshold for the maximum percentage of unique mitochondrial gene identifiers detected in a single cell.

`verbose` Print progress

Value

return an assay containing predicted expression value in the data slot

---

SaveHeatmap *This function will take a set of Seurat objects and plot them as a heat map. Normalization options include no normalization - raw counts, log1p, sctransform, pflog1ppf*

Description

This function will take a set of Seurat objects and plot them as a heat map. Normalization options include no normalization - raw counts, log1p, sctransform, pflog1ppf

Usage

```
SaveHeatmap(cell.data, reference.data, n.plot, glist, save.dir, verbose = TRUE)
```

Arguments

`cell.data`	A Seurat object pre-processed with aocseq::CombineData.
`reference.data`	A reference matrix.
`n.plot`	Number of cells in heatmap.
`glist`	Character array. Gene name, a marker of interest.
`save.dir`	Character array. Gene name, a marker of interest.
`verbose`	Print progress.

Value

Void.

`SegmentPlot` *Functions to plot data stored in clonal objects at normalized counts greater than the control*

Description

Functions to plot data stored in clonal objects at normalized counts greater than the control

Usage

```
SegmentPlot(
  SummaryTable,
  names = c("S1", "S2"),
  segment.names = c("S1", "S2"),
  n.segments = 50,
  tune.gap = 0.005,
  segment.names.spacer = "16mm",
  percentage.spacer.1 = "14mm",
  percentage.spacer.2 = "13mm",
  segment.col = c("green", "blue"),
  ribbon.hue = "pink",
  save.dir = "circos.pdf"
)
```

Arguments

`SummaryTable`	An aocseq summary table listing cell type frequencies.
`names`	Names of samples that are being compared.
`n.segments`	Number of segments to plot.
`tune.gap`	Resolution of circos plot.
`segment.names.spacer`	Parameter to control spacing of circos plot labels.
`percentage.spacer.1`	Parameter to control spacing of circos plot labels of sample 1.
`percentage.spacer.2`	Parameter to control spacing of circos plot labels of sample 2.

`segment.col`	Colors of circos plot labels.
`ribbon.hue`	Colors of circos plot ribbons.
`save.dir`	save.dir to output file.

Value

A circos plot of aocseq cell type frequencies.

`SetCellType` *This function sets the cell.types as meta data.*

Description

This function sets the cell.types as meta data.

Usage

```
SetCellType(cell.data, cell.types)
```

Arguments

`cell.data`	A Seurat Object.
`cell.types`	Single cell metadata.

Value

A Seurat Object.

`UMAPReduce` *Plot UMAP dimension reduction with aocseq meta data.*

Description

This function will save dimension reduction UMAP plots generated by Seurat and overlay aocseq metadata.

Usage

```
UMAPReduce(
  cell.data,
  save.dir = ".",
  cell.types = c(),
  ident.list = c("orig.ident"),
  rseed = 356,
  ur.nfeatures = 500,
  preload = FALSE,
  preloadsave.dir = "."
)
```

Arguments

`cell.data`	A Seurat object pre-processed with aocseq::CombineData.
`save.dir`	save.dir to output file.
`cell.types`	List of cell types for UMAP annotation.
`ident.list`	Character array. Gene name, a marker of interest.
`rseed`	Set seed for data recall.
`ur.nfeatures`	Choose fewer dimension reduction parameters for rapid visualization of aocseq cell type metadata.
`preload`	True or False. If integration anchors must be loaded, set this parameter to True, otherwise they will be computed.
`preloadsave.dir`	save.dir to preload data, if preload is set to True.
`verbose`	Print progress.

Value

return a Seurat UMAP plot with aocseq annotation. slot

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